

GED Mathematics Worksheet- Rational Expressions and Functions

Instructions

Solve all 40 problems. Show your work where applicable. Simplify all answers completely.

Part 1: Multiplying Rational Expressions (Questions 1-10)

1. Multiply: $\frac{3x}{4} \cdot \frac{8}{9x}$
2. Multiply: $\frac{5a^2}{6b} \cdot \frac{12b}{10a}$
3. Multiply: $\frac{x+2}{x-1} \cdot \frac{x-1}{x+3}$
4. Multiply: $\frac{2x^2}{5y} \cdot \frac{15y^2}{8x}$
5. Multiply: $\frac{(x+4)(x-3)}{(x+1)} \cdot \frac{(x+5)}{(x-3)}$
6. Multiply: $\frac{x^2-4}{x^2+5x+6} \cdot \frac{x+3}{x-2}$
7. Multiply: $\frac{6m^2n}{7} \cdot \frac{14}{18mn^2}$
8. Multiply: $\frac{x^2-9}{x^2-1} \cdot \frac{x+1}{x+3}$
9. Multiply: $\frac{a^2-a-6}{a^2+4a+3} \cdot \frac{a+1}{a-3}$
10. Multiply: $\frac{2x^2-8x}{x^2-16} \cdot \frac{x+4}{2x}$

Part 2: Dividing Rational Expressions (Questions 11-20)

11. Divide: $\frac{5}{8} \div \frac{3}{4}$
12. Divide: $\frac{x^2}{y} \div \frac{x}{2y}$

13. Divide: $\frac{3a}{5b} \div \frac{9a^2}{10b^2}$
14. Divide: $\frac{m^2n}{6} \div \frac{mn^2}{12}$
15. Divide: $\frac{x+2}{x-3} \div \frac{x+2}{x+5}$
16. Divide: $\frac{x^2-4}{x^2+6x+9} \div \frac{x-2}{x+3}$
17. Divide: $\frac{x^2-8x}{x^2+8x+16} \div \frac{x-8}{x^2-2x-24}$
18. Divide: $\frac{2x^2+5x+3}{x^2-1} \div \frac{2x+3}{x-1}$
19. Divide: $\frac{a^2-9}{a^2-4} \div \frac{a+3}{a-2}$
20. Divide: $\frac{x^2-5x+6}{x^2+x-12} \div \frac{x-2}{x+4}$

Part 3: Understanding Functions (Questions 21-30)

21. Determine if the relation $\{(1, 2), (2, 4), (3, 6), (4, 8)\}$ is a function. Explain why or why not.
22. Determine if the relation $\{(1, 3), (2, 5), (1, 7), (3, 9)\}$ is a function. Explain why or why not.
23. For $f(x) = 3x - 2$, find $f(5)$.
24. For $g(x) = x^2 + 4$, find $g(-3)$.
25. For $h(x) = 2x^2 - 5x + 1$, find $h(2)$.
26. For $f(x) = \frac{x+3}{x-1}$, find $f(4)$.
27. For $g(x) = 5x - 7$, find $g(-2)$.
28. For $h(x) = x^3 - 2x + 4$, find $h(3)$.
29. For $f(x) = \sqrt{x+5}$, find $f(4)$.
30. For $g(x) = |2x - 3|$, find $g(1)$.

Part 4: Evaluating Functions with Expressions (Questions 31-35)

31. For $g(x) = 3x + 5$, find $g(4n)$.
32. For $h(x) = 2x^2 - 3x + 7$, find $h(4y^2)$.
33. For $f(x) = x^2 - 6x$, find $f(a + 2)$.

34. For $g(x) = 5x - 1$, find $g(2m + 3)$.
35. For $h(x) = x^2 + 2x - 8$, find $h(x - 1)$.

Part 5: Functions and Graphs (Questions 36-40)

36. Use the vertical line test to determine if the graph of $y = x^2$ represents a function.
37. Use the vertical line test to determine if the graph of $x = y^2$ represents a function.
38. For the function $f(x) = 2x - 3$, find $f(2)$ using the equation. Then verify your answer by substituting into the function.
39. Describe the behavior of a function during the interval $-2 \leq x < 1$ if the function increases throughout this interval.
40. For a piecewise function, explain why different intervals might have different behaviors (increasing, decreasing, or constant).

Answer Key

Part 1: Multiplying Rational Expressions

1. $\frac{2}{3}$
2. a
3. 1 (with restrictions: $x \neq 1, -3$)
4. $\frac{3xy}{2}$
5. $\frac{(x+4)(x+5)}{x+1}$ or $\frac{x^2+9x+20}{x+1}$
6. $\frac{x+2}{1}$ or $x+2$ (with restrictions: $x \neq -3, -2, 2$)
7. $\frac{4m}{3n}$
8. $\frac{x-3}{x-1}$ (with restrictions: $x \neq 1, -1, 3$)
9. $\frac{a-2}{1}$ or $a-2$ (with restrictions: $a \neq -3, -1, 3$)
10. $x-2$ (with restrictions: $x \neq 0, 4, -4$)

Part 2: Dividing Rational Expressions

11. $\frac{5}{6}$
12. $2y$
13. $\frac{2b}{3a}$
14. $2m$
15. $\frac{x+5}{x-3}$ (with restrictions: $x \neq -2, 3, -5$)
16. $\frac{x+3}{1}$ or $x+3$ (with restrictions: $x \neq -3, 2$)
17. $\frac{x(x-6)}{x+4}$ (with restrictions: $x \neq 0, -4, 8$)
18. $\frac{x+1}{1}$ or $x+1$ (with restrictions: $x \neq 1, -1$)
19. $\frac{a-3}{a-2}$ (with restrictions: $a \neq 3, -3, 2$)
20. $\frac{x-3}{1}$ or $x-3$ (with restrictions: $x \neq 2, -4$)

Part 3: Understanding Functions

21. Yes, it is a function. Each domain element (1, 2, 3, 4) is paired with exactly one range element.
22. No, it is not a function. The domain element 1 is paired with two different range elements (3 and 7).
23. $f(5) = 13$
24. $g(-3) = 13$
25. $h(2) = -1$
26. $f(4) = \frac{7}{3}$
27. $g(-2) = -17$
28. $h(3) = 25$
29. $f(4) = 3$
30. $g(1) = 1$

Part 4: Evaluating Functions with Expressions

31. $g(4n) = 12n + 5$
32. $h(4y^2) = 32y^4 - 12y^2 + 7$
33. $f(a + 2) = a^2 - 2a - 8$
34. $g(2m + 3) = 10m + 14$
35. $h(x - 1) = x^2 - 4x - 5$

Part 5: Functions and Graphs

36. Yes, $y = x^2$ represents a function. Any vertical line intersects the parabola at most once, so it passes the vertical line test.
37. No, $x = y^2$ does not represent a function. A vertical line at $x = 4$ (for example) intersects the parabola at two points, so it fails the vertical line test.
38. Using the equation: $f(2) = 2(2) - 3 = 4 - 3 = 1$. Verification: Substituting $x = 2$ into $f(x) = 2x - 3$ gives $f(2) = 1$.
39. During the interval $-2 \leq x < 1$, the function increases as x increases. This means that as you move from left to right along the graph in this interval, the y -values get larger.

40. Different intervals can have different behaviors because a piecewise function is defined by different rules on different intervals. For example, one part might follow a linear rule (constant slope), another might follow a quadratic rule (changing slope), and another might be constant (zero slope).